WORKED SOLUTION

# STEP III Specimen paper, Question 10

Pure • Modular Arithmetic

**STEP III Specimen paper, Question 10**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q10

## The key idea

Both parts are about remainders, so work modulo a well-chosen number. In (i), squares can only take a few values modulo 8, and three of those values can never add up to 7. In (ii), rewrite the powers so that both terms involve the same base modulo 17; the two coefficients then add up to a multiple of 17.

## Solution

### Part (i)

*Odd squares.* Write an odd number as $2k+1$, where $k$ is an integer. Then

$$(2k+1)^2 = 4k^2 + 4k + 1 = 4k(k+1) + 1.$$

Of the two consecutive integers $k$ and $k+1$, one is even, so $k(k+1)$ is even and $4k(k+1)$ is a multiple of 8. Hence every odd square leaves remainder 1 on division by 8.

*Even squares.* Write an even number as $2k$. Then $(2k)^2 = 4k^2$.

- If $k$ is even, $k = 2j$, then $4k^2 = 16j^2$, a multiple of 8: remainder 0.
- If $k$ is odd, then $k^2$ is odd, say $k^2 = 2j+1$, so $4k^2 = 8j+4$: remainder 4.

So every even square leaves remainder 0 or 4 on division by 8.

*The deduction.* Every square is congruent to 0, 1 or 4 modulo 8 (and this includes $0^2 = 0$, so we are allowing some of the three squares to be zero). Suppose $N = a^2 + b^2 + c^2$. Then $N$ mod 8 is the remainder of a sum of three numbers each chosen from $\{0, 1, 4\}$. Sorting by how many of them equal 4:

| number of 4s | possible sums of the rest, and the total mod 8 |
|---|---|
| 0 | $0, 1, 2, 3$ |
| 1 | $4 + \{0, 1, 2\} = 4, 5, 6$ |
| 2 | $8 + \{0, 1\} \equiv 0, 1$ |
| 3 | $12 \equiv 4$ |

The possible remainders are $0, 1, 2, 3, 4, 5, 6$, and $7$ never appears. A number of the form $8n + 7$ leaves remainder 7 on division by 8, so it cannot be a sum of three squares.

**Ask yourself** Why can't we just check that 7 itself isn't a sum of three squares?

Because the claim is about infinitely many numbers $7, 15, 23, \ldots$, and checking them one by one never finishes. Working modulo 8 handles them all at once: whatever $a$, $b$, $c$ are, only their remainders matter, and the table covers every case.

#### Part (ii)

Split off the powers so that each term is a constant times an $n$th power:

$$2^{3n+1} = 2 \cdot 8^n, \qquad 3\left(5^{2n+1}\right) = 15 \cdot 25^n.$$

Now $25 = 17 + 8$, so $25 \equiv 8 \pmod{17}$, and therefore $25^n \equiv 8^n \pmod{17}$ for every $n \geq 0$ (congruences can be multiplied). Hence

$$\begin{aligned} 2^{3n+1} + 3\left(5^{2n+1}\right) &= 2 \cdot 8^n + 15 \cdot 25^n \\ &\equiv 2 \cdot 8^n + 15 \cdot 8^n \\ &= 17 \cdot 8^n \equiv 0 \pmod{17}. \end{aligned}$$

So 17 divides $2^{3n+1} + 3(5^{2n+1})$ for all integers $n \geq 0$, as required.

**Ask yourself** What if I haven't met congruences?

The same idea works by induction. Let $f(n) = 2 \cdot 8^n + 15 \cdot 25^n$. Then $f(0) = 17$, and

$$f(n+1) - 8f(n) = 15 \cdot 25^{n+1} - 15 \cdot 8 \cdot 25^n = 15 \cdot 25^n(25 - 8) = 17 \cdot 15 \cdot 25^n,$$

so $f(n+1) = 8f(n) + 17 \cdot 15 \cdot 25^n$. If $17 \mid f(n)$ then $17 \mid f(n+1)$, and the result follows by induction.

## Checks

Try small cases. $7 = 4 + 1 + 1 + 1$ needs four squares, and $15 = 9 + 4 + 1 + 1$ also needs four. On the other hand $6 = 4 + 1 + 1$ and $14 = 9 + 4 + 1$ are fine. A computer search confirms that no number $8n + 7$ up to 5000 is a sum of three squares. For (ii): $n = 0$ gives $2 + 15 = 17$, and $n = 1$ gives $16 + 375 = 391 = 17 \times 23$. The expression was checked to be divisible by 17 for every $n$ from 0 to 299.

## Common slips

- Forgetting that 0 is a square, so a "sum of three squares" may really use only one or two non-zero squares. The argument above includes this case automatically, since 0 is one of the residues.
- Saying odd squares are 1 mod 8 because "$4k^2 + 4k$ is divisible by 4": that only gives 1 mod 4. You need the factor of 2 from $k(k+1)$.

## Beyond STEP

Part (i) is the easy half of Legendre's three-square theorem: a positive integer is a sum of three squares *exactly* when it is not of the form $4^a(8n+7)$. The converse direction is much harder, and needs the theory of quadratic forms. By contrast Lagrange proved that *every* positive integer is a sum of four squares. Part (ii) is the everyday arithmetic of $\mathbb{Z}/17\mathbb{Z}$, which first-year courses develop into the study of groups and rings. Davenport's *The Higher Arithmetic* covers congruences and sums of squares beautifully.

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